Coding Assignment 2

Due: 6 November, 2026 6pm PT

Minimum Length Path on a Sorted Graph

Problem Description

Consider a graph with N vertices, labelled 1 through N, and M directed edges, where each edge is associated with a length ℓ. The edges of the graph all satisfy a special property: the edges are of the form $u \xrightarrow{\ell} v$ where $u < v$.

Write a program that prints the minimum length of a path from 1 to N. It is guaranteed that there exists a path from 1 to N.

Input

The input has the following form.

N	M	
u_1	v_1	ℓ_1
u_2	v_2	ℓ_2
$\ldots$		

Specifically, the input consists of $M + 1$ lines:

- The first line of the input contains two integers N ($5 \leq N \leq 2 \cdot 10^5$) and M ($8 \leq M \leq 10^6$).
- Each of the next M lines describes an edge of the graph with three integers: u_i and v_i, the two endpoints of the edge ($1 \leq u_i < v_i \leq N$); and w_i ($-10^5 \leq w_i \leq 10^5$), the length of the edge.
  - Note: For this problem, all edges will satisfy $u_i < v_i$.

Output

Output 1 line with a single integer, the path from 1 to N with minimum length. If no such path exists, print -1.

Constraints

- $5 \leq N \leq 2 \cdot 10^5$
- $8 \leq M \leq 10^6$
- $1 \leq u_i < v_i \leq N$

- $-10^5 \leq w_i \leq 10^5$
- The time limit is 1.5 seconds for C/C++ and 4 seconds for Python.

Test Cases

Your program will be evaluated on 5 visible test cases and 5 hidden test cases. Each test case is worth 0.6 points.

The 5 visible test cases are in a single zip file (`CA2.zip`) on Canvas.

Sample Input 1:

```
5 8
1 4 2
1 3 8
1 5 10
2 5 6
1 2 10
2 4 7
4 5 9
3 5 2
```

Sample Output 1:

```
10
```

Sample Input 2:

```
6 12
1 2 12
1 5 17
4 6 20
1 4 -3
2 4 12
3 6 16
5 6 0
2 5 14
1 3 20
2 6 9
3 4 14
3 5 -6
```

Sample Output 2:

```
14
```

Sample Explanation

The minimum length path in Sample 1 is $1 \to 3 \to 5$, with total length $8 + 2 = 10$.

The minimum length path in Sample 2 is $1 \to 3 \to 5 \to 6$, with total length $20 - 6 + 0 = 14$.

Submission Guideline

Write your program in either C, C++ or Python **in a single file**. Name your file `GraphPath.ext` where `ext` is `c`, `cpp` or `py` depending on your language. Submit the file on Gradescope. You can make at most 10 submission attempts.